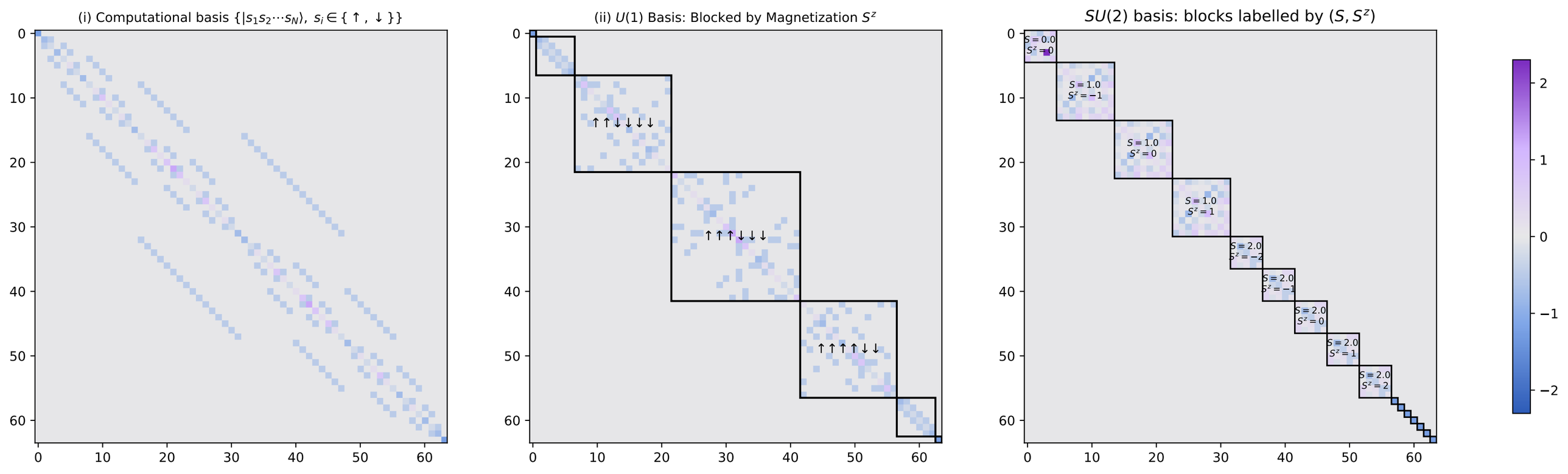
(i) Computational basis $\{|s_1 s_2 \cdots s_N\rangle,\ s_i \in \{\uparrow, \downarrow\}\}$
(ii) $U(1)$ Basis: Blocked by Magnetization S^z
$SU(2)$ basis: blocks labelled by (S, S^z)
$\uparrow\uparrow\downarrow\downarrow\downarrow$
$\uparrow\uparrow\uparrow\downarrow\downarrow$
$\uparrow\uparrow\uparrow\uparrow\downarrow\downarrow$
$S = 0.0$ $S^z = 0$
$S = 1.0$ $S^z = -1$
$S = 1.0$ $S^z = 0$
$S = 1.0$ $S^z = 1$
$S = 2.0$ $S^z = -2$
$S = 2.0$ $S^z = -1$
$S = 2.0$ $S^z = 0$
$S = 2.0$ $S^z = 1$
$S = 2.0$ $S^z = 2$